

- **Topic:** Relational Algebra
- Assignments must be completed individually.
- No late assignments will be accepted.
- When writing Relational Algebra expressions, follow the formatting style in the lectures. If you write your answers without proper formatting, it will be difficult to understand your answers. This will increase the chances of your answers being misinterpreted, and not earning full points.
- The questions here are based on the Enrollment database whose schema is listed on the last page. Write relational algebra expressions to answer the following questions.
- Submit your completed solutions to **Crowdmark**.
- **Rubric:** Each question is worth 3 points, and will be graded according to the following rubric:
 - 3 points: Good (no mistake or only a very small mistake)
 - 2 points: OK (mostly correct)
 - 1 point: Poor (mostly wrong but has some merit)
 - 0 points: Wrong (totally off or no answer)

[3]

1. Find students who **never** took more than one course from the same instructor. That is, every course that they took (including repeated courses if any) had a different instructor. The result should list student name and ID.

* For every student-instructor pairs, we count # of courses, this includes the case of repeated courses

$R1 \leftarrow \pi_{SID, InstructorID} \left(\sigma_{count(course\#) \leq 1} (Enroll) \right)$ as num-courses

* we filter R1 and find students that violate the more than 1 course per instructor rule.

$R2 \leftarrow \pi_{SID} \left(\sigma_{num-courses > 1} (R1) \right)$

$AS \leftarrow \pi_{SID} (Enroll)$

* we do a set subtraction, to get the students that do not violate the rule.

$NS \leftarrow \underbrace{AS}_{\text{All students}} - \underbrace{R2}_{\text{violation students}}$

$RESULT \leftarrow \pi_{Name, SID} (NS * Student)$

whyd we have to do set subtraction, why cant we just filter for num courses ≤ 1 ?

great question. think about what's in Enroll vs Student —

a student who never enrolled in anything wouldn't appear in Enroll at all, so they'd never show up in R1 after the aggregation. but they still satisfy the condition (they never took two courses with the same instructor).

set subtraction from `$\pi_{SID}(Enroll)$` is fine here since the question implicitly assumes students who took courses. but if you filtered R1 for num-courses ≤ 1 , you'd miss anyone with exactly one instructor total who only appears once — actually wait, that would be caught.

the real issue is: **students who only ever had one enrollment per instructor are already in R1 with count = 1**, so filtering ≤ 1 would work on R1. but students with zero enrollments aren't in R1 at all.

so both approaches have the same gap. the set subtraction is just the more standard "find the bad ones, remove them" pattern that's cleaner to reason about. your way would also work for the students captured in R1!

good rule of thumb:

use **find bad ones, subtract** when the condition is "never" / "for all" / "no case where X happened." these are hard to express as a direct filter because the absence of something doesn't show up as a row.

use **direct filter** when the condition is a simple property of a row or group — like "grade > 50" or "count > 5."

the tell is usually the word "**never**" or "**every**" in the question. Q1 said "never took more than one course from the same instructor" — that's an absence condition, so subtract. if it had said "find students with at most 3 enrollments" you could just aggregate and filter directly.

- [3] 2. Find course(s) offered in W26 for which all the students were from the same major, or the course used textbook(s) published by "Pearson Publishing". The result should display course name and course number.

$$WC \leftarrow \pi_{\text{course\#}} \left(\sigma_{\text{Term} = 'W26'} (\text{Enroll}) \right)$$

$$C \leftarrow \pi_{\text{course\#}} \left(\sigma_{\text{Publisher} = \text{"Pearson Publishing"} \text{ AND } \text{Term} = 'W26'} (\text{Textbook} * \text{BOOK}) \right)$$

$$R \leftarrow \sigma_{\text{Term} = 'W26'} (\text{Enroll} * \text{Student})$$

$$R1 \leftarrow \text{course\#} \int \text{count-distinct(major) as num-majors} (R)$$

$$R2 \leftarrow \sigma_{\text{num-majors} > 1} (R1)$$

$$SM \leftarrow \pi_{\text{course\#}} (WC - R2)$$

$$I \leftarrow C \cup SM$$

$$\text{Result} \leftarrow \pi_{\text{course\#, cname}} (I * \text{course})$$

 **ASK.**

[3]

3. In the Math Faculty, a student can attempt the same course at most three times. We need to give a friendly reminder to students who are attempting the same course for the third time this term (S26). Write a query to find these students. The result should display their name, SID and major. Assume that the policy has now been adopted by all Faculties and applied to all students.

3rd attempt - ① (SID, Course#) appears exactly 3 times in enroll
 ② one of those appearances has term = 'S26'

$R1 \leftarrow \pi_{SID, Course\#} \rho_{count(Term) \text{ as num-attempts}} (Enroll)$

$R2 \leftarrow \pi_{SID, Course\#} (\sigma_{num-attempts = 3} (R1))$

$R3 \leftarrow \pi_{SID} (\sigma_{Term = 'S26'} (R2 * Enroll))$

$Result \leftarrow \pi_{SID, Name, major} (R3 * Student)$

[3]

4. Find the publisher(s) who has at least one book adopted by each department except the AMATH department (i.e. ZERO books have been adopted by courses offered by AMATH department).

$$R1 \leftarrow (Course * (Book * Textbook))$$

joins on Course #
gives ISBN, Title, Publisher, Author, Course#, Term
Cname, Dept

joins on ISBN:
ISBN, Title, Publisher, Author, Course#, Term.

$$R2 \leftarrow \pi_{\text{Publisher, Dept}}(R1)$$

our dividend.

$$D \leftarrow \sigma_{\text{Dept} \neq \text{'AMATH'}}(\pi_{\text{Dept}}(\text{course}))$$

*All non-Amath departments → our divisor.

$$R3 \leftarrow R2 \div D$$

left with publishers who appear with every department in D.

$$R4 \leftarrow \pi_{\text{Publisher}}(\sigma_{\text{Dept} = \text{'AMATH'}}(R1))$$

Publisher with books used by AMATH.

$$\text{Result} \leftarrow R3 - R4$$

↓
could have all departments

remove any who show up in AMATH.

division vs set subtraction:

- **division** — "for every X, this entity has X" — the entity must cover a complete set.
"publishers who have a book in every department"
- **set subtraction** — "this entity never has X" — removing entities that have an unwanted property.
"students who never took two courses with the same instructor"

the tell: "every/all" → division. "never/no/except" → subtraction.

note your Q4 actually has both — you used filtering (implicit subtraction of AMATH) AND division together. the AMATH condition is handled by just excluding it upfront before dividing.

if you have $R(A, B, C) \div S(C)$, a tuple (a, b) survives if for **every** c in S , the tuple (a, b, c) exists in R . the "key" that survives is everything in R **except** the divisor columns, and it must pair with every value in the divisor.

so with more columns you'd get more specific survivors — a (Publisher, SomeOtherCol) pair would need to cover every dept, not just Publisher alone. that's why you project down to just (Publisher, Dept) before dividing — you only want Publisher to be the surviving part.

$R2 \div D$: a publisher survives if for **every** dept in D , the (publisher, dept) pair exists in $R2$. extra depts (like AMATH) in $R2$ are just **ignored** by division — they don't help or hurt.

so yes, a publisher with all departments including AMATH survives the division, and then gets knocked out by $- R4$. that's exactly why you need the subtraction step. your solution handles it correctly!

for your general confusion — the rule is: publisher P survives $\div D$ if $\{P\} \times D \subseteq R2$. anything outside D in $R2$ is irrelevant.

[3]

5. Find out which of the following majors has the most students: C&O, AMATH or ITM. The result should be displayed in the provided format below. The first row contains the program names. The second row contains the number of students in each program. For example,

C&O	AMATH	ITM
35	67	31

Hint: Use the cross-product.

$$SM \leftarrow \text{Major } \underbrace{g_{\text{count}(SID) \text{ as num-students}} \left(\pi_{SID, \text{Major}} (Student) \right)}_{\text{get counts for every major}}$$

$$M1(C\&O) \leftarrow \pi_{\text{num-students}} \left(\sigma_{\text{Major} = 'C\&O'} (SM) \right)$$

gives us relation M1 with column C&O that contains the number of students.

$$M2(AMATH) \leftarrow \pi_{\text{num-students}} \left(\sigma_{\text{Major} = 'AMATH'} (SM) \right)$$

$$M3(ITM) \leftarrow \pi_{\text{num-students}} \left(\sigma_{\text{Major} = 'ITM'} (SM) \right)$$

$$\text{Result} \leftarrow \underbrace{M1 \times (M2 \times M3)}_{\text{cross product to combine all rows}}$$

[3]

6. Find the publisher(s) who has at least 10 different books adopted by courses offered by the CS department, but NONE adopted by the STAT department.

$$R1 \leftarrow (Course * (Book * Textbook))$$

joins on Course #
gives
ISBN, Title, Publisher,
Author, Course#, Term
Cname, Dept

joins on ISBN:
ISBN, Title, Publisher, Author,
Course#, Term.

$$R2 \leftarrow \text{Publisher } \rho_{\text{count-distinct}(\text{ISBN}) \text{ as num-books } (\sigma_{\text{dept}='CS'}(R1))}$$

Since we care about
different books

$$R3 \leftarrow \pi_{\text{Publisher}} (\sigma_{\text{dept}='STAT'}(R1))$$

Publishers with books in STAT department

$$R4 \leftarrow \pi_{\text{Publisher}} (\sigma_{\text{num-books} \geq 10}(R2))$$

Publishers with atleast 10 different books
adopted by CS courses.

$$\text{Result} \leftarrow R4 - R3$$

- [3] 7. Find students who have earned at least 5 credits (in 2A or beyond) but have yet to pass the first communication requirement. Assume each course is 0.5 credit and the only courses in the first communication requirement are ENGL109 or SPCOM100. Note that a student might have taken the same course twice and passed both times. This counts as 0.5 credit, not 1 credit. Say John took MATH135 twice and got 50 and 65. John earned 0.5 credit from MATH135. Jane took MATH135 and got 55 and is currently taking it again in F24. Jane has also earned 0.5 credit from MATH135. The result should display student name, SID and E-mail.

All passing enrollments

$R1 \leftarrow \sigma_{\text{Grade} \geq 50 \text{ and Grade} \neq 'NA' \text{ and Grade} \neq 'WD'} (Enroll)$

$R2 \leftarrow \pi_{SID} \rho_{\text{count-distinct (course\#)} \text{ as num-courses}} (R1)$

Students with ≥ 10 distinct passing courses (at least 5 credits)

$R3 \leftarrow \pi_{SID} (\sigma_{\text{num-courses} \geq 10} (R2))$

Students who passed communication requirement

$R4 \leftarrow \pi_{SID} (\sigma_{\text{course\#} = 'SPCOM100' \text{ OR course\#} = 'ENGL109'} (R1))$

$Result1 \leftarrow R3 - R4$

$Result \leftarrow \pi_{\text{Name, SID, Email}} (Result1 * Student)$

- [3] 8. For pairs of students who are in the same major, find the pair(s) who took classes together (same course in the same term) the most frequently (across all majors). Note that (A, B) and (B, A) is the same pair, and only one should be displayed. The result should show the SID.

Enroll1 $\leftarrow \Pi_{SID, course\#, major, Term} (Enroll * Student)$

Enroll2 $\leftarrow P_{Enroll2(SID2, Course\#, Term, Major)}(Enroll1)$

$R1 \leftarrow \sigma_{SID < SID2} (Enroll1 * Enroll2)$

$\downarrow$
 we select distinct pairs

natural join matches on course#, term, major.

$R2 \leftarrow \underbrace{SID, SID2 \ \rho \ \text{count}(course\#) \text{ as num-together}}_{(R1)}$

for each pair we count how many times it appeared together.

$R3 \leftarrow \underbrace{\rho \ \text{max}(\text{num-together}) \text{ as max-together}}_{(R2)}$

get the maximum frequency.

$R4 \leftarrow \underbrace{R2 \bowtie_{num-together = max-together} R3}_{(R3)}$

get pair with the maximum frequency

Result $\leftarrow \Pi_{SID, SID2} (R4)$

sure! so in R1 you have:

- Enroll1: (SID, Course#, Term, Major)
- Enroll2: (SID2, Course#, Term, Major)

natural join matches on all shared column names — which are Course#, Term, Major. so a row in R1 means "SID and SID2 were in the same Course#, same Term, and same Major." that's exactly the condition you need!

the $\sigma(\text{SID} < \text{SID2})$ on top just eliminates duplicate pairs.

the earlier confusion was me suggesting a theta join — but natural join works perfectly here because you renamed SID to SID2 in Enroll2, so SID doesn't get joined on (no shared name), but Course#, Term, Major do. that's the key insight — **renaming SID was intentional so it wouldn't be joined on, while keeping Course#/Term/Major shared so they would be.**

- [3] 9. Find the Math/BA student(s) with the second highest average. The result should display their name, SID and major.

$R1 \leftarrow \pi_{SID, GRADE} \left(\sigma_{Major = "Math/BA"} (Enroll * Student) \right)$

$C1 \leftarrow \sigma_{Grade \neq 'NA' \text{ AND } Grade \neq 'WD'} (R1)$
 Filter out invalid grades before calculating average.

$R2 \leftarrow \pi_{SID} \rho_{avg(GRADE) \text{ as } avg-grade} (C1)$
 Compute student averages.

$R3 \leftarrow \rho_{max(avg-grade) \text{ as } max-avg-grade} (R2)$

$NHG1 \leftarrow \pi_{SID} \bowtie_{avg-grade \neq max-avg-grade} R2 \quad R3$
 removes "top" students.

$R4 \leftarrow \rho_{max(avg-grade) \text{ as } second-avg-grade} (NHG1)$

$R5 \leftarrow \pi_{SID} \left(\bowtie_{avg-grade = second-avg-grade} R2 \quad R4 \right)$
 students whose average is second highest

$Result \leftarrow \pi_{Name, SID, Major} (R5 * Student)$

- [3] 10. Find the most difficult course(s) in W26. The difficulty of a course is estimated by a weighted factoring formula:

$$(\text{Number of failures} * 0.6 + \text{Number of WD} * 0.2 - \text{Class Average} * 0.2).$$

The higher this number, the more difficult the course is.

Hint: You can use arithmetic in the projection operator (π) as follows:

Say we have $R(AB)$, $R1(C) \leftarrow \pi_{A-B}(R)$. This produces a new relation R1 having a column C. Each row in C is the difference between A and B in R.

The result should display the Course#.

$$R1 \leftarrow \pi_{\text{Course\#}, \text{Grade}} \left(\sigma_{\text{Grade} \neq 'NA' \text{ AND TERM} = 'W26'} (Enroll) \right)$$

$$F \leftarrow \sigma_{\text{Grade} < 50 \text{ AND Grade} \neq 'WD'} (R1)$$

$$FC \leftarrow \text{Course\#} \text{ } g \text{ count(grade) as num-fails } (F)$$

$$WD \leftarrow \sigma_{\text{Grade} = 'WD'} (R1)$$

$$WDC \leftarrow \text{Course\#} \text{ } g \text{ count(grade) as num-wd } (WD)$$

$$NWD \leftarrow \sigma_{\text{Grade} \neq 'WD'} (R1)$$

$$A \leftarrow \text{course\#} \text{ } g \text{ avg(grade) as avg-grade } (NWD)$$

$$V \leftarrow FC \bowtie (WDC \bowtie A)$$

outer join in case course didnt have fails or wd

$$R(C, D) \leftarrow \pi_{\text{course\#}, (\text{num-fails} * 0.6) + (\text{num-wd} * 0.2) - (\text{avg-grade} * 0.2)} (V)$$

$$DC \leftarrow g \text{ max(D) as max-d } R$$

$$\text{Result} \leftarrow \pi_{\text{Course\#}} (R \bowtie_{D = \text{max-d}} DC)$$

The Enrollment Database

The following relations for a database keep track of student enrollment in courses and the books adopted for each course.

Assumptions:

- Examples of Course#: CS338, MATH135;
- Examples of Dept: CS, ACTSC, AMATH, ENGL;
- Examples of Major: CS, Math/BA, SE, STAT;
- Examples of Term: F24, W24;
- For students who are taking a course but have yet to receive a grade, the grade will be recorded as NA;
- For students who enroll into a course but drop it midway through, the grade will be recorded as WD;
- A grade below 50 is a failure;
- Publisher is unique.

Relations:

- Student(SID, Name, Major, E-mail, Bdate)
- Course(Course#, Cname, Dept)
- Enroll(SID, Course#, Term, Grade, InstructorID)
- Textbook(Course#, Term, ISBN)
- Book(ISBN, Title, Publisher, Author)

Note that in some question(s), you might consider using the **Count–Distinct(a)** aggregation function instead of **Count(a)**. The former counts the number of **distinct** values in attribute a.